RELATIVE DE RHAM POINCARE DUALITY AS A PERFECT PAIRING OF PERFECT COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. Original author theorem and proof are retained. Only this target is counted; local supporting statements are not separate new entries.

Allowed standard prerequisites (editorial).

- Derived categories, perfect complexes and duality, relative Hodge-to-de Rham spectral sequences, relative coherent cohomological dimension and Serre duality. The local relative Hodge and boundary-degeneration arguments are retained.. These are the named standard inputs; the selected main construction and its local nonroutine support remain in full.

Source reference. Original section 07HX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9567–9567. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original section 0FKU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19474–19474. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0G8G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20266–20266. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0G7Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20248–20248. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original section 03GX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4544–4544. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 01UX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2426–2426. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 02GU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3241–3241. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original remark 0B6S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14080–14080. Stable Stacks locator for declared

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standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0BRT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14781–14781. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0FM0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19515–19515. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original remark 0G81: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20251–20251. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 01ZM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2599–2599. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0C0C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15082–15082. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0EY2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18677–18677. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 07RR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9876–9876. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 081F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10215–10215. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0H99: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21519–21519. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0FW6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19836–19836. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 00EW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 550–550. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 00EU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 548–548. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 02KH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3372–3372. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original theorem 03H2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4549–4549. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0G80: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20250–20250. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 01F6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1859–1859. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 01XC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2513–2513. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 055U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6687–6687. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original section 01KT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2062–2062. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 063K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7834–7834. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original section 067R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7980–7980. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original remark 0G8A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20260–20260. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0E7D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17778–17778. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0G8D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20263–20263. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0G8E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20264–20264. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0FMA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19525–19525. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0FLQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19505–19505. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0FU4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19764–19764. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0G8B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20261–20261. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original section 0FM9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19524–19524. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original section 0G4A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20120–20120. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0G8C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20262–20262. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original remark 0G6W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20211–20211. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original remark 0G7W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20246–20246. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0BCG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14279–14279. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original proposition 0FW7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19837–19837. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original section 0FM6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19521–19521. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0FW4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19834–19834. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 014A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1465–1465. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

1. Cup product

Consider the maps $\Omega^p_{X/S} \times \Omega^q_{X/S} \to \Omega^{p+q}_{X/S}$ given by $(\omega, \eta) \longmapsto \omega \wedge \eta$. Using the formula for d given in Section 07HX and the Leibniz rule for $\mathrm{d} : \mathcal{O}_X \to \Omega_{X/S}$ we see that $\mathrm{d}(\omega \wedge \eta) = \mathrm{d}(\omega) \wedge \eta + (-1)^{\deg(\omega)} \omega \wedge \mathrm{d}(\eta)$. This means that $\wedge$ defines a morphism

$$\wedge : \mathrm{Tot}(\Omega^\bullet_{X/S} \otimes_{p^{-1}\mathcal{O}_S} \Omega^\bullet_{X/S}) \longrightarrow \Omega^\bullet_{X/S} \tag{1.0.1}$$

of complexes of $p^{-1}\mathcal{O}_S$-modules.

Combining the cup product of Cohomology, Section 0FKU with (1.0.1) we find a $H^0(S, \mathcal{O}_S)$-bilinear cup product map

$$\cup : H^i_{dR}(X/S) \times H^j_{dR}(X/S) \longrightarrow H^{i+j}_{dR}(X/S)$$

For example, if $\omega \in \Gamma(X, \Omega^i_{X/S})$ and $\eta \in \Gamma(X, \Omega^j_{X/S})$ are closed, then the cup product of the de Rham cohomology classes of ω and η is the de Rham cohomology class of $\omega \wedge \eta$, see discussion in Cohomology, Section 0FKU.

Given a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{f} & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

of schemes, the pullback maps $f^* : R\Gamma(X, \Omega^\bullet_{X/S}) \to R\Gamma(X', \Omega^\bullet_{X'/S'})$ and $f^* : H^i_{dR}(X/S) \longrightarrow H^i_{dR}(X'/S')$ are compatible with the cup product defined above.

Lemma 1.1. *In Situation 0G8G the pushforward $f_*\mathcal{O}_X$ is a finite étale $\mathcal{O}_S$-algebra and locally on S we have $Rf_*\mathcal{O}_X = f_*\mathcal{O}_X \oplus P$ in $D(\mathcal{O}_S)$ with P perfect of tor amplitude in $[1, \infty)$. The map $d : f_*\mathcal{O}_X \to f_*\Omega_{X/S}$ is zero.*

Proof. The first part of the statement follows from Derived Categories of Schemes, Lemma 0G7Y. Setting $S' = \underline{\mathrm{Spec}}_S(f_*\mathcal{O}_X)$ we get a factorization $X \to S' \to S$ (this is the Stein factorization, see More on Morphisms, Section 03GX, although we don't need this) and we see that $\Omega_{X/S} = \Omega_{X/S'}$ for example by Morphisms, Lemma 01UX and 02GU. This of course implies that $\mathrm{d} : f_*\mathcal{O}_X \to f_*\Omega_{X/S}$ is zero. □

Lemma 1.2. *In Situation 0G8G there exists an $\mathcal{O}_S$-module map*

$$t : Rf_*\Omega^n_{X/S}[n] \longrightarrow \mathcal{O}_S$$

unique up to precomposing by multiplication by a unit of $H^0(X, \mathcal{O}_X)$ with the following property: for all p the pairing

$$Rf_*\Omega^p_{X/S} \otimes^{\mathbf{L}}_{\mathcal{O}_S} Rf_*\Omega^{n-p}_{X/S}[n] \longrightarrow \mathcal{O}_S$$

given by the relative cup product composed with t is a perfect pairing of perfect complexes on S.

Proof. We first assume S is a Noetherian scheme. Let $\omega^\bullet_{X/S}$ be the relative dualizing complex of X over S as in Duality for Schemes, Remark 0B6S and let $Rf_*\omega^\bullet_{X/S} \to \mathcal{O}_S$ be its trace map. By Duality for Schemes, Lemma 0BRT (here we use that S is Noetherian) there exists an isomorphism $\omega^\bullet_{X/S} \cong \Omega^n_{X/S}[n]$ and using this isomorphism we obtain t. The complexes $Rf_*\Omega^p_{X/S}$ are perfect by Lemma 0FM0. Since $\Omega^p_{X/S}$ is locally free and since $\Omega^p_{X/S} \otimes_{\mathcal{O}_X} \Omega^{n-p}_{X/S} \to \Omega^n_{X/S}$ exhibits an isomorphism $\Omega^p_{X/S} \cong \mathcal{H}om_{\mathcal{O}_X}(\Omega^{n-p}_{X/S}, \Omega^n_{X/S})$ we see that the pairing induced by the relative cup product is perfect by Duality for Schemes, Remark 0G81.

Proof of existence in the general case. By absolute Noetherian approximation we can find a cartesian diagram

$$\begin{array}{ccc} X & \longrightarrow & X_0 \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f_0} \\ S & \xrightarrow{g} & S_0 \end{array}$$

with S_0 Noetherian and f_0 a proper and smooth morphism all of whose fibres are nonempty and equidimensional of dimension n. See Limits, Lemmas 01ZM, 0C0C, 0EY2, 07RR, and 081F. Choose a map

$$t_0 : Rf_{0,*}\Omega^n_{X_0/S_0}[n] \longrightarrow \mathcal{O}_{S_0}$$

as in the statement of the lemma. By cohomology and base change the complexes $Rf_*(\Omega^p_{X/S})$ are equal to $Lg^*(Rf_{0,*}\Omega^p_{X_0/S_0})$ for all $p \geq 0$, see Lemma 0FM0. In particular the pullback of t_0 by g gives us a map t as in the statement of the lemma. Namely, by Cohomology, Lemma 0H99, the cup product map

$$Rf_*\Omega^p_{X/S} \otimes^{\mathbf{L}}_{\mathcal{O}_S} Rf_*\Omega^{n-p}_{X/S}[n] \longrightarrow Rf_*\Omega^n_{X/S}$$

is the pullback by g of the corresponding cup product map for f_0. Hence the fact that t_0 produces perfect pairings, pulls back to the same property for t on S.

Uniqueness of t. Choose a distinguished triangle $f_*\mathcal{O}_X \to Rf_*\mathcal{O}_X \to P \to f_*\mathcal{O}_X[1]$. By Lemma 1.1 the object P is perfect of tor amplitude in $[1, \infty)$ and the triangle is locally on S split. Thus $R\mathcal{H}om_{\mathcal{O}_X}(P, \mathcal{O}_X)$ is perfect of tor amplitude in $(-\infty, -1]$. Hence duality (above) shows that locally on S we have

$$Rf_*\Omega^n_{X/S}[n] \cong R\mathcal{H}om_{\mathcal{O}_S}(f_*\mathcal{O}_X, \mathcal{O}_S) \oplus R\mathcal{H}om_{\mathcal{O}_X}(P, \mathcal{O}_X)$$

This shows that $R^n f_*\Omega^n_{X/S}$ is finite locally free and that we obtain a perfect $\mathcal{O}_S$-bilinear pairing

$$f_*\mathcal{O}_X \times R^n f_*\Omega^n_{X/S} \longrightarrow \mathcal{O}_S$$

using t. This implies that any $\mathcal{O}_S$-linear map $t' : R^n f_*\Omega^n_{X/S} \to \mathcal{O}_S$ is of the form $t' = t \circ g$ for some $g \in \Gamma(S, f_*\mathcal{O}_X) = \Gamma(X, \mathcal{O}_X)$. In order for t' to still determine a perfect pairing g will have to be a unit. This finishes the proof. □

Lemma 1.3. *In Situation 0G8G the map $d : R^n f_*\Omega^{n-1}_{X/S} \to R^n f_*\Omega^n_{X/S}$ is zero.*

As we mentioned in the proof of Lemma 0FW6 this lemma is not an easy consequence of Lemmas 1.2 and 1.1.

Proof in case S is reduced. Assume S is reduced. Observe that $\mathrm{d} : R^n f_* \Omega^{n-1}_{X/S} \to R^n f_* \Omega^n_{X/S}$ is an $\mathcal{O}_S$-linear map of (quasi-coherent) $\mathcal{O}_S$-modules. The $\mathcal{O}_S$-module $R^n f_* \Omega^n_{X/S}$ is finite locally free (as the dual of the finite locally free $\mathcal{O}_S$-module $f_* \mathcal{O}_X$ by Lemmas 1.2 and 1.1). Since S is reduced it suffices to show that the stalk of d in every generic point $\eta \in S$ is zero; this follows by looking at sections over affine opens, using that the target of d is locally free, and Algebra, Lemma 00EW part (2). Since S is reduced we have $\mathcal{O}_{S,\eta} = \kappa(\eta)$, see Algebra, Lemma 00EU. Thus d_η is identified with the map

$$\mathrm{d} : H^n(X_\eta, \Omega^{n-1}_{X_\eta/\kappa(\eta)}) \longrightarrow H^n(X_\eta, \Omega^n_{X_\eta/\kappa(\eta)})$$

which is zero by Lemma 0FW6. □

Proof in the general case. Observe that the question is flat local on S: if $S' \to S$ is a surjective flat morphism of schemes and the map is zero after pullback to S', then the map is zero. Also, formation of the map commutes with base change by flat morphisms by flat base change (Cohomology of Schemes, Lemma 02KH).

Consider the Stein factorization $X \to S' \to S$ as in More on Morphisms, Theorem 03H2. By Lemma 1.1 the morphism $\pi : S' \to S$ is finite étale. The morphism $f : X \to S'$ is proper (by the theorem), smooth (by More on Morphisms, Lemma 0G80) with geometrically connected fibres by the theorem on Stein factorization. In the proof of Lemma 1.1 we saw that $\Omega_{X/S} = \Omega_{X/S'}$ because $S' \to S$ is étale. Hence $\Omega^\bullet_{X/S} = \Omega^\bullet_{X/S'}$. We have

$$R^q f_* \Omega^p_{X/S} = \pi_* R^q f'_* \Omega^p_{X/S'}$$

for all p, q by the Leray spectral sequence (Cohomology, Lemma 01F6), the fact that π is finite hence affine, and Cohomology of Schemes, Lemma 01XC (of course we also use that $R^q f'_* \Omega^p_{X'/S}$ is quasi-coherent). Thus the map of the lemma is π_* applied to $\mathrm{d} : R^n f'_* \Omega^{n-1}_{X/S'} \to R^n f'_* \Omega^n_{X/S'}$. In other words, in order to prove the lemma we may replace $f : X \to S$ by $f' : X \to S'$ to reduce to the case discussed in the next pargraph.

Assume f has geometrically connected fibres and $f_* \mathcal{O}_X = \mathcal{O}_S$. For every $s \in S$ we can choose an étale neighbourhood $(S', s') \to (S, s)$ such that the base change $X' \to S'$ of S has a section. See More on Morphisms, Lemma 055U. By the initial remarks of the proof this reduces us to the case discussed in the next paragraph.

Assume f has geometrically connected fibres, $f_* \mathcal{O}_X = \mathcal{O}_S$, and we have a section $s : S \to X$ of f. We may and do assume $S = \mathrm{Spec}(A)$ is affine. The map $s^* : R\Gamma(X, \mathcal{O}_X) \to R\Gamma(S, \mathcal{O}_S) = A$ is a splitting of the map $A \to R\Gamma(X, \mathcal{O}_X)$. Thus we can write

$$R\Gamma(X, \mathcal{O}_X) = A \oplus P$$

where P is the "kernel" of s^*. By Lemma 1.1 the object P of $D(A)$ is perfect of tor amplitude in $[1, n]$. As in the proof of Lemma 1.2 we see that $H^n(X, \Omega^n_{X/S})$ is a locally free A-module of rank 1 (and in fact dual to A so free of rank 1 – we will soon choose a generator but we don't want to check it is the same generator nor will it be necessary to do so).

Denote $Z \subset X$ the image of s which is a closed subscheme of X by Schemes, Lemma 01KT. Observe that $Z \to X$ is a regular (and a fortiori Koszul regular by Divisors, Lemma 063K) closed immersion by Divisors, Lemma 067R. Of course $Z \to X$ has codimension n. Thus by Remark 0G8A we can consider the map

$$\gamma^{0,0} : H^0(Z, \Omega^0_{Z/S}) \longrightarrow H^n(X, \Omega^n_{X/S})$$

and we set $\xi = \gamma^{0,0}(1) \in H^n(X, \Omega^n_{X/S})$.

We claim ξ is a basis element. Namely, since we have base change in top degree (see for example Limits, Lemma 0E7D) we see that $H^n(X, \Omega^n_{X/S}) \otimes_A k = H^n(X_k, \Omega^n_{X_k/k})$ for any ring map $A \to k$. By the compatibility of the construction of ξ with base change, see Lemma 0G8D, we see that the image of ξ in $H^n(X_k, \Omega^n_{X_k/k})$ is nonzero by Lemma 0G8E if k is a field. Thus ξ is a nowhere vanishing section of an invertible module and hence a generator.

Let $\theta \in H^n(X, \Omega^{n-1}_{X/S})$. We have to show that $\mathrm{d}(\theta)$ is zero in $H^n(X, \Omega^n_{X/S})$. We may write $\mathrm{d}(\theta) = a\xi$ for some $a \in A$ as ξ is a basis element. Then we have to show $a = 0$.

Consider the closed immersion

$$\Delta : X \to X \times_S X$$

This is also a section of a smooth morphism (namely either projection) and hence a regular and Koszul immersion of codimension n as well. Thus we can consider the maps

$$\gamma^{p,q} : H^q(X, \Omega^p_{X/S}) \longrightarrow H^{q+n}(X \times_S X, \Omega^{p+n}_{X \times_S X/S})$$

of Remark 0G8A. Consider the image

$$\gamma^{n-1,n}(\theta) \in H^{2n}(X \times_S X, \Omega^{2n-1}_{X \times_S X})$$

By Lemma 0FMA we have

$$\Omega^{2n-1}_{X \times_S X} = \Omega^{n-1}_{X/S} \boxtimes \Omega^n_{X/S} \oplus \Omega^n_{X/S} \boxtimes \Omega^{n-1}_{X/S}$$

By the Künneth formula (either Derived Categories of Schemes, Lemma 0FLQ or Derived Categories of Schemes, Lemma 0FU4) we see that

$$H^{2n}(X \times_S X, \Omega^{n-1}_{X/S} \boxtimes \Omega^n_{X/S}) = H^n(X, \Omega^{n-1}_{X/S}) \otimes_A H^n(X, \Omega^n_{X/S})$$

and

$$H^{2n}(X \times_S X, \Omega^n_{X/S} \boxtimes \Omega^{n-1}_{X/S}) = H^n(X, \Omega^n_{X/S}) \otimes_A H^n(X, \Omega^{n-1}_{X/S})$$

Namely, since we are looking in top degree there no higher tor groups that intervene. Combined with the fact that ξ is a generator this means we can write

$$\gamma^{n-1,n}(\theta) = \theta_1 \otimes \xi + \xi \otimes \theta_2$$

with $\theta_1, \theta_2 \in H^n(X, \Omega^{n-1}_{X/S})$. Arguing in exactly the same manner we can write

$$\gamma^{n,n}(\xi) = b\xi \otimes \xi$$

in $H^{2n}(X \times_S X, \Omega^{2n}_{X \times_S X/S}) = H^n(X, \Omega^n_{X/S}) \otimes_A H^n(X, \Omega^n_{X/S})$ for some $b \in H^0(S, \mathcal{O}_S)$.

Claim: $\theta_1 = \theta$, $\theta_2 = \theta$, and $b = 1$. Let us show that the claim implies the desired result $a = 0$. Namely, by Lemma 0G8B we have

$$\gamma^{n,n}(\mathrm{d}(\theta)) = \mathrm{d}(\gamma^{n-1,n}(\theta))$$

By our choices above this gives

$$a\xi \otimes \xi = \gamma^{n,n}(a\xi) = \mathrm{d}(\theta \otimes \xi + \xi \otimes \theta) = a\xi \otimes \xi + (-1)^n a\xi \otimes \xi$$

The right most equality comes from the fact that the map $\mathrm{d} : \Omega^{2n-1}_{X \otimes_S X/S} \to \Omega^{2n}_{X \times_S X/S}$ by Lemma 0FMA is the sum of the differential $\mathrm{d} \boxtimes 1 : \Omega^{n-1}_{X/S} \boxtimes \Omega^n_{X/S} \to \Omega^n_{X/S} \boxtimes \Omega^n_{X/S}$ and the differential $(-1)^n 1 \boxtimes \mathrm{d} : \Omega^n_{X/S} \boxtimes \Omega^{n-1}_{X/S} \to \Omega^n_{X/S} \boxtimes \Omega^n_{X/S}$. Please see discussion in Section 0FM9 and Derived Categories of Schemes, Section 0G4A for more information. Since $\xi \otimes \xi$ is a basis for the rank 1 free A-module $H^n(X, \Omega^n_{X/S}) \otimes_A H^n(X, \Omega^n_{X/S})$ we conclude

$$a = a + (-1)^n a \Rightarrow a = 0$$

as desired.

In the rest of the proof we prove the claim above. Let us denote $\eta = \gamma^{0,0}(1) \in H^n(X \times_S X, \Omega^n_{X\times_S X/S})$. Since $\Omega^n_{X\times_S X/S} = \bigoplus_{p+p'=n} \Omega^p_{X/S} \boxtimes \Omega^{p'}_{X/S}$ we may write

$$\eta = \eta_0 + \eta_1 + \ldots + \eta_n$$

where η_p is in $H^n(X \times_S X, \Omega^p_{X/S} \boxtimes \Omega^{n-p}_{X/S})$. For $p = 0$ we can write

$$\begin{aligned} H^n(X \times_S X, \mathcal{O}_X \boxtimes \Omega^n_{X/S}) &= H^n(R\Gamma(X, \mathcal{O}_X) \otimes^{\mathbf{L}}_A R\Gamma(X, \Omega^n_{X/S})) \\ &= A \otimes_A H^n(X, \Omega^n_{X/S}) \oplus H^n(P \otimes^{\mathbf{L}}_A R\Gamma(X, \Omega^n_{X/S})) \end{aligned}$$

by our previously given decomposition $R\Gamma(X, \mathcal{O}_X) = A \oplus P$. Consider the morphism $(s, \mathrm{id}) : X \to X \times_S X$. Then $(s, \mathrm{id})^{-1}(\Delta) = Z$ scheme theoretically. Hence we see that $(s, \mathrm{id})^*\eta = \xi$ by Lemma 0G8D. This means that

$$\xi = (s, \mathrm{id})^*\eta = (s^* \otimes \mathrm{id})(\eta_0)$$

This means exactly that the first component of η_0 in the direct sum decomposition above is ξ. In other words, we can write

$$\eta_0 = 1 \otimes \xi + \eta_0'$$

with $\eta_0' \in H^n(P \otimes^{\mathbf{L}}_A R\Gamma(X, \Omega^n_{X/S}))$. In exactly the same manner for $p = n$ we can write

$$\begin{aligned} H^n(X \times_S X, \Omega^n_{X/S} \boxtimes \mathcal{O}_X) &= H^n(R\Gamma(X, \Omega^n_{X/S}) \otimes^{\mathbf{L}}_A R\Gamma(X, \mathcal{O}_X)) \\ &= H^n(X, \Omega^n_{X/S}) \otimes_A A \oplus H^n(R\Gamma(X, \Omega^n_{X/S}) \otimes^{\mathbf{L}}_A P) \end{aligned}$$

and we can write

$$\eta_n = \xi \otimes 1 + \eta_n'$$

with $\eta_n' \in H^n(R\Gamma(X, \Omega^n_{X/S}) \otimes^{\mathbf{L}}_A P)$.

Observe that $\mathrm{pr}_1^*\theta = \theta \otimes 1$ and $\mathrm{pr}_2^*\theta = 1 \otimes \theta$ are Hodge cohomology classes on $X \times_S X$ which pull back to θ by Δ. Hence by Lemma 0G8C we have

$$\theta_1 \otimes \xi + \xi \otimes \theta_2 = \gamma^{n-1,n}(\theta) = (\theta \otimes 1) \cup \eta = (1 \otimes \theta) \cup \eta$$

in the Hodge cohomology ring of $X \times_S X$ over S. In terms of the direct sum decomposition on the modules of differentials of $X \times_S X/S$ we obtain

$$\theta_1 \otimes \xi = (\theta \otimes 1) \cup \eta_0 \quad \text{and} \quad \xi \otimes \theta_2 = (1 \otimes \theta) \cup \eta_n$$

Looking at the formula $\eta_0 = 1 \otimes \xi + \eta_0'$ we found above, we see that to show that $\theta_1 = \theta$ it suffices to prove that

$$(\theta \otimes 1) \cup \eta_0' = 0$$

To do this, observe that cupping with $\theta \otimes 1$ is given by the action on cohomology of the map

$$(P \otimes^{\mathbf{L}}_A R\Gamma(X, \Omega^n_{X/S}))[-n] \xrightarrow{\theta\otimes 1} R\Gamma(X, \Omega^{n-1}_{X/S}) \otimes^{\mathbf{L}}_A R\Gamma(X, \Omega^n_{X/S})$$

in the derived category, see Cohomology, Remark 0G6W. This map is the derived tensor product of the two maps

$$\theta : P[-n] \to R\Gamma(X, \Omega^{n-1}_{X/S}) \quad \text{and} \quad 1 : R\Gamma(X, \Omega^n_{X/S}) \to R\Gamma(X, \Omega^n_{X/S})$$

by Derived Categories of Schemes, Remark 0G7W. However, the first of these is zero in $D(A)$ because it is a map from a perfect complex of tor amplitude in $[n+1, 2n]$ to a complex with cohomology only in degrees $0, 1, \ldots, n$, see More on Algebra, Lemma 0BCG. A similar argument works to show the vanishing of $(1 \otimes \theta) \cup \eta_n'$. Finally, in exactly the same manner we obtain

$$b\xi \otimes \xi = \gamma^{n,n}(\xi) = (\xi \otimes 1) \cup \eta_0$$

and we conclude as before by showing that $(\xi \otimes 1) \cup \eta_0' = 0$ in the same manner as above. This finishes the proof. □

Proposition 1.4. *Let S be a quasi-compact and quasi-separated scheme. Let $f : X \to S$ be a proper smooth morphism of schemes all of whose fibres are nonempty and equidimensional of dimension n. There exists an $\mathcal{O}_S$-module map*

$$t : R^{2n} f_* \Omega^\bullet_{X/S} \longrightarrow \mathcal{O}_S$$

unique up to precomposing by multiplication by a unit of $H^0(X, \mathcal{O}_X)$ with the following property: the pairing

$$Rf_*\Omega^\bullet_{X/S} \otimes^{\mathbf{L}}_{\mathcal{O}_S} Rf_*\Omega^\bullet_{X/S}[2n] \longrightarrow \mathcal{O}_S, \quad (\xi, \xi') \longmapsto t(\xi \cup \xi')$$

is a perfect pairing of perfect complexes on S.

Proof. The proof is exactly the same as the proof of Proposition 0FW7.

By the relative Hodge-to-de Rham spectral sequence

$$E_1^{p,q} = R^q f_* \Omega^p_{X/S} \Rightarrow R^{p+q} f_* \Omega^\bullet_{X/S}$$

(Section 0FM6), the vanishing of $\Omega^i_{X/S}$ for $i > n$, the vanishing in for example Limits, Lemma 0E7D and the results of Lemmas 1.1 and 1.3 we see that $R^0 f_* \Omega_{X/S} = R^0 f_* \mathcal{O}_X$ and $R^n f_* \Omega^n_{X/S} = R^{2n} f_* \Omega^\bullet_{X/S}$. More precisely, these identifications come from the maps of complexes

$$\Omega^\bullet_{X/S} \to \mathcal{O}_X[0] \quad \text{and} \quad \Omega^n_{X/S}[-n] \to \Omega^\bullet_{X/S}$$

Let us choose $t : R^{2n} f_* \Omega_{X/S} \to \mathcal{O}_S$ which via this identification corresponds to a t as in Lemma 1.2.

Let us abbreviate $\Omega^\bullet = \Omega^\bullet_{X/S}$. Consider the map (1.0.1) which in our situation reads

$$\wedge : \mathrm{Tot}(\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \Omega^\bullet) \longrightarrow \Omega^\bullet$$

For every integer $p = 0, 1, \ldots, n$ this map annihilates the subcomplex $\mathrm{Tot}(\sigma_{>p}\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p}\Omega^\bullet)$ for degree reasons. Hence we find that the restriction of $\wedge$ to the subcomplex $\mathrm{Tot}(\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \geq_{n-p} \Omega^\bullet)$ factors through a map of complexes

$$\gamma_p : \mathrm{Tot}(\sigma_{\leq p}\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p}\Omega^\bullet) \longrightarrow \Omega^\bullet$$

Using the same procedure as in Section 1 we obtain relative cup products

$$Rf_*\sigma_{\leq p}\Omega^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_S} Rf_*\sigma_{\geq n-p}\Omega^\bullet \longrightarrow Rf_*\Omega^\bullet$$

We will prove by induction on p that these cup products via t induce perfect pairings between $Rf_*\sigma_{\leq p}\Omega^\bullet$ and $Rf_*\sigma_{\geq n-p}\Omega^\bullet[2n]$. For $p = n$ this is the assertion of the proposition.

The base case is $p = 0$. In this case we have

$$Rf_*\sigma_{\leq p}\Omega^\bullet = Rf_*\mathcal{O}_X \quad \text{and} \quad Rf_*\sigma_{\geq n-p}\Omega^\bullet[2n] = Rf_*(\Omega^n[-n])[2n] = Rf_*\Omega^n[n]$$

In this case we simply obtain the pairing between $Rf_*\mathcal{O}_X$ and $Rf_*\Omega^n[n]$ of Lemma 1.2 and the result is true.

Induction step. Say we know the result is true for p. Then we consider the distinguished triangle

$$\Omega^{p+1}[-p-1] \to \sigma_{\leq p+1}\Omega^\bullet \to \sigma_{\leq p}\Omega^\bullet \to \Omega^{p+1}[-p]$$

and the distinguished triangle

$$\sigma_{\geq n-p}\Omega^\bullet \to \sigma_{\geq n-p-1}\Omega^\bullet \to \Omega^{n-p-1}[-n+p+1] \to (\sigma_{\geq n-p}\Omega^\bullet)[1]$$

Observe that both are distinguished triangles in the homotopy category of complexes of sheaves of $f^{-1}\mathcal{O}_S$-modules; in particular the maps $\sigma_{\leq p}\Omega^\bullet \to \Omega^{p+1}[-p]$

and $\Omega^{n-p-1}[-d+p+1] \to (\sigma_{\geq n-p}\Omega^\bullet)[1]$ are given by actual maps of complexes, namely using the differential $\Omega^p \to \Omega^{p+1}$ and the differential $\Omega^{n-p-1} \to \Omega^{n-p}$. Consider the distinguished triangles associated gotten from these distinguished triangles by applying Rf_*

$$\begin{array}{ccc}
Rf_*\sigma_{\leq p}\Omega^\bullet & & Rf_*\sigma_{\geq n-p}\Omega^\bullet \\
\downarrow{\scriptstyle a} & & \uparrow{\scriptstyle a'} \\
Rf_*\Omega^{p+1}[-p-1] & & Rf_*\Omega^{n-p-1}[-n+p+1] \\
\downarrow{\scriptstyle b} & & \uparrow{\scriptstyle b'} \\
Rf_*\sigma_{\leq p+1}\Omega^\bullet & & Rf_*\sigma_{\geq n-p-1}\Omega^\bullet \\
\downarrow{\scriptstyle c} & & \uparrow{\scriptstyle c'} \\
Rf_*\sigma_{\leq p}\Omega^\bullet & & Rf_*\sigma_{\geq n-p}\Omega^\bullet \\
\downarrow{\scriptstyle d} & & \uparrow{\scriptstyle d'} \\
Rf_*\Omega^{p+1}[-p-1] & & Rf_*\Omega^{n-p-1}[-n+p+1]
\end{array}$$

We will show below that the pairs (a, a'), (b, b'), (c, c'), and (d, d') are compatible with the given pairings. This means we obtain a map from the distinguished triangle on the left to the distinguished triangle obtained by applying $R\mathcal{H}om(-, \mathcal{O}_S)$ to the distinguished triangle on the right. By induction and Lemma 0FW4 we know that the pairings constructed above between the complexes on the first, second, fourth, and fifth rows are perfect, i.e., determine isomorphisms after taking duals. By Derived Categories, Lemma 014A we conclude the pairing between the complexes in the middle row is perfect as desired.

Let $e : K \to K'$ and $e' : M' \to M$ be maps of objects of $D(\mathcal{O}_S)$ and let $K \otimes^{\mathbf{L}}_{\mathcal{O}_S} M \to \mathcal{O}_S$ and $K' \otimes^{\mathbf{L}}_{\mathcal{O}_S} M' \to \mathcal{O}_S$ be pairings. Then we say these pairings are compatible if the diagram

$$\begin{array}{ccc}
K' \otimes^{\mathbf{L}}_{\mathcal{O}_S} M' & \xleftarrow[e\otimes 1]{} & K \otimes^{\mathbf{L}}_{\mathcal{O}_S} M' \\
\downarrow & & \downarrow{\scriptstyle 1\otimes e'} \\
\mathcal{O}_S & \longleftarrow & K \otimes^{\mathbf{L}}_{\mathcal{O}_S} M
\end{array}$$

commutes. This indeed means that the diagram

$$\begin{array}{ccc}
K & \longrightarrow & R\mathcal{H}om(M, \mathcal{O}_S) \\
\downarrow{\scriptstyle e} & & \downarrow{\scriptstyle R\mathcal{H}om(e', -)} \\
K' & \longrightarrow & R\mathcal{H}om(M', \mathcal{O}_S)
\end{array}$$

commutes and hence is sufficient for our purposes.

Let us prove this for the pair (c, c'). Here we observe simply that we have a commutative diagram

$$\begin{array}{ccc}
\mathrm{Tot}(\sigma_{\leq p}\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p}\Omega^\bullet) & \longleftarrow & \mathrm{Tot}(\sigma_{\leq p+1}\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p}\Omega^\bullet) \\
\downarrow{\scriptstyle \gamma_p} & & \downarrow \\
\Omega^\bullet & \xleftarrow{\gamma_{p+1}} & \mathrm{Tot}(\sigma_{\leq p+1}\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p-1}\Omega^\bullet)
\end{array}$$

By functoriality of the cup product we obtain commutativity of the desired diagram.

Similarly for the pair (b, b') we use the commutative diagram

$$\begin{array}{ccc} \mathrm{Tot}(\sigma_{\leq p+1}\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p-1}\Omega^\bullet) & \longleftarrow & \mathrm{Tot}(\Omega^{p+1}[-p-1] \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p-1}\Omega^\bullet) \\ \downarrow{\scriptstyle \gamma_{p+1}} & & \downarrow \\ \Omega^\bullet & \xleftarrow{\wedge} & \Omega^{p+1}[-p-1] \otimes_{f^{-1}\mathcal{O}_S} \Omega^{n-p-1}[-n+p+1] \end{array}$$

For the pairs (d, d') and (a, a') we use the commutative diagram

$$\begin{array}{ccc} \Omega^{p+1}[-p] \otimes_{f^{-1}\mathcal{O}_S} \Omega^{n-p-1}[-n+p] & \longleftarrow & \mathrm{Tot}(\sigma_{\leq p}\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \Omega^{n-p-1}[-n+p]) \\ \downarrow & & \downarrow \\ \Omega^\bullet & \longleftarrow & \mathrm{Tot}(\sigma_{\leq p}\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p}\Omega^\bullet) \end{array}$$

We omit the argument showing the uniqueness of t up to precomposing by multiplication by a unit in $H^0(X, \mathcal{O}_X)$. $\square$